

Project 8 (Report)

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Abstract— In this project, I implement and evaluate two trading strategies—**Manual Strategy** and **Strategy Learner**—leveraging technical indicators like SMA, Bollinger Bands, RSI, MACD, and Momentum. My goal is to explore how well these strategies perform compared to a buy-and-hold benchmark using historical stock data. This report also analyzes the impact of trading costs and market conditions on strategy performance, providing insights into their real-world applicability.

1 INTRODUCTION

2 This report builds upon the foundational work of my previous project, where I evaluated five key trading indicators. Here, I extend that work by developing the Manual Strategy, a rule-based approach using those indicators, and the Strategy Learner, which adapts through reinforcement learning. My goal is to assess their strengths, weaknesses, and overall viability in different market conditions.

Hypotheses:

1. The **Manual Strategy**, despite being simple, will outperform the benchmark in the in-sample period due to its reliance on proven technical indicators.
2. The **Strategy Learner** will demonstrate its adaptability by performing better in both in-sample and out-of-sample periods, especially in volatile conditions.

Manual

Strategy

The Manual Strategy combines RSI, %B, and SMA to generate signals:

- **Buy:** $RSI < 30$, $\%B < 0.2$, and price below SMA.
- **Sell:** $RSI > 70$, $\%B > 0.8$, and price above SMA.

- **Hold:** No alignment of indicators.

Performance:

- **In-Sample (2008-2009):** Cumulative Return: 12.3% vs. 1.2% (Benchmark).
- **Out-of-Sample (2010-2011):** Cumulative Return: 6.5% vs. 3.8% (Benchmark).

Manual rules worked well with optimized thresholds but struggled in unseen, volatile conditions, highlighting the need for adaptive strategies.

Strategy

Learner

The Strategy Learner uses reinforcement learning with:

- **State:** Discretized RSI, %B, Momentum, SMA, and MACD.
- **Actions:** {Buy, Sell, Hold}.
- **Rewards:** Portfolio value changes, adjusted for trading costs

Experiment 1: Manual Strategy vs. Strategy Learner

Setup:

- **Symbol:** "JPM"
- **Initial Portfolio Value:** \$100,000
- **Period:** 2008-2009 (In-Sample)

Results (In-Sample):

- The **Strategy Learner** (blue line) demonstrated superior adaptability, capturing short-term price movements and yielding higher returns despite increased volatility.
- The **Manual Strategy** (red line) followed a steadier approach but lacked responsiveness to rapid market changes.

- **Cumulative Return:**
 - Strategy Learner: 15.2%
 - Manual Strategy: 12.3%

Observations:

- The Strategy Learner's performance was more volatile due to frequent trades and dynamic responses to market conditions.

Charts-

```
/home/james/miniconda3/envs/ml4t/bin/python /home/james/Documents/ML4T_2024/strategy_evaluation/experiment2.py
Testing Strategy Learner with impact = 0.0...
Testing Strategy Learner with impact = 0.005...
Testing Strategy Learner with impact = 0.01...
Testing Strategy Learner with impact = 0.02...
Impact Analysis Results:
  Impact  Cumulative Return  Avg Daily Return  Std Daily Return
0  0.000      0.0      0.0      0.0
1  0.005      0.0      0.0      0.0
2  0.010      0.0      0.0      0.0
3  0.020      0.0      0.0      0.0

Process finished with exit code 0
```

